CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 8: Logarithms

Every formula and result of the chapter on one sheet

1	Definition	$a^x = N \iff \log_a N = x$
2	Log of 1	$\log_a 1 = 0$
3	Log of the base	$\log_a a = 1$
4	Log of a power of the base	$\log_a a^n = n$
5	Inverse	$a^{\log_a N} = N$
6	Product law	$\log_a (mn) = \log_a m + \log_a n$
7	Quotient law	$\log_a \frac{m}{n} = \log_a m - \log_a n$
8	Power law	$\log_a m^n = n \log_a m$
9	Root	$\log_a \sqrt[n]{m} = \frac{1}{n} \log_a m$
10	Equal logs	$\log_a m = \log_a n$ gives $m = n$
11	Common logs	$\log 10 = 1, \log 100 = 2, \log 0.1 = -1$
12	From $\log 2$	$\log 5 = 1 - \log 2$